

Tower Of Hanoi
–Alternative Way of Calculation of Time Complexity

$$\text{We have } T(n) = 2T(n - 1) + 1 ,$$

Where, $2T(n - 1)$ is Homogeneous Part and $+1$ is Non Homogenous Part.

Hence , we cannot make $T(n) = x^n$ directly, due to Non – Homogenous part is present.

To cancel out $+ 1$, Let take : $T(n) = S(n) - C$

Applying it to : $T(n) = 2T(n - 1) + 1$, we get:

$$S(n) - C = 2(S(n - 1) - C) + 1$$

$$S(n) - C = 2S(n - 1) - 2C + 1$$

$$S(n) = 2S(n - 1) - 2C + C + 1$$

$$S(n) = 2S(n - 1) - C + 1$$

Now, to cancel out $+ 1$:

$$-C + 1 = 0$$

$$\Rightarrow -C = -1$$

$$\Rightarrow C = 1$$

$$\therefore S(n) = 2S(n - 1) - 1 + 1$$

$$\therefore S(n) = 2S(n - 1)$$

Method 1: Substitution

$$\text{Now, } S(n - 1) = 2S(n - 2)$$

Substituting $S(n - 1) = 2S(n - 2)$ in $T(n)$ we get:

$$S(n) = 2[2S(n - 2)]$$

$$\Rightarrow 2^2 S(n - 2)$$

$$\text{Again, } S(n - 2) = 2S(n - 3)$$

Substituting $T(n - 2) = 2T(n - 3)$ in $T(n)$ we get:

$$T(n) = 2^2 S(n - 2)$$

$$\Rightarrow 2^2 [2S(n - 3)]$$

$$\Rightarrow 2^3 S(n - 3)$$

Now, if it runs upto `k` times we get:

$$S(n) = 2^k S(n - k)$$

As, our base case is $T(0)$ not $T(1)$, hence:

$n - k = 0$, therefore, $k = n$, we get:

$$S(n) = 2^n S(n - n)$$

$$\Rightarrow 2^n S(0)$$

$$\text{Let, } A = S(0)$$

$$S(n) = A \times 2^n$$

Method 2: Assumption

$$\text{Assume: } S(n) = x^n$$

$$x^n = 2x^{n-1}$$

Dividing x^{n-1} in both the sides we get:

$$\frac{x^n}{x^{n-1}} = \frac{2x^{n-1}}{x^{n-1}}$$

$$\Rightarrow x = 2$$

So:

$$S(n) = x^n \Rightarrow S(n) \propto 2^n \text{ [proportional to } 2^n \text{]}$$

If 2^n is a solution then:

$A \times 2^n$ is also a solution.

Where, A is a constant.

$$S(n) \propto 2^n \Rightarrow S(n) = A \times 2^n \left[\begin{array}{l} \text{As, we are using } A \text{ as} \\ \text{a constant, then } \propto \\ \text{turns } =. \end{array} \right]$$

That is ,If $S(n) \propto f(n)$, then $S(n) = A \times f(n)$

“Proportional ($\propto$) always expands to equality with a constant multiplier.”

We know,

$$T(n) = S(n) - C \text{ and } C = 1, \text{ Hence, } T(n) = S(n) - 1$$

$$\therefore S(n) = T(n) + 1$$

$$\Rightarrow S(0) = T(0) + 1$$

$$\Rightarrow A = 1 + 1 [T(0) = 1 \text{ and } S(0) = A]$$

$$\Rightarrow A = 2$$

$$\therefore S(n) = 2 \times 2^n$$

$$\therefore S(n) = 2^{n+1}$$

$$\therefore T(n) = 2^{n+1} - 1$$

putting Big O we get:

$$\Rightarrow O(2^{n+1} - 1)$$

$$\Rightarrow O(2^{n+1}) - O(1)$$

$$\Rightarrow O(2^{n+1})$$

$$\Rightarrow O(2^n \times 2)$$

$$\Rightarrow O(2^n) \times O(2)$$

$$\Rightarrow O(2^n \times 2)$$

$$\Rightarrow O(2^n).$$

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